Solutions to FIT3080 Applied Class on Introduction to Machine Learning

Exercise 1: Supervised and unsupervised learning

(a) Identifying species of flora and fauna is more of an exercise in supervised learning or unsupervised learning?

SOLUTION:
Unsupervised learning – as we are just identifying groups of things for the first time.

(b) Learning to recognize species of flora and fauna that have already been classified is more of an exercise in supervised learning or unsupervised learning?

SOLUTION:
Supervised learning – as we are trying to match items to labels.

(c) You want to train a supervised Machine Learning algorithm for the following tasks, what will be your labelled data?

i. To help you predict how long will it take for you to drive home from your workplace.
SOLUTION:
Note: It is possible to have multiple answers, so long as the data are relevant, e.g.,
Weather conditions
Time of the day
Holidays
Route
Label: actual time it took

ii. To help you recognize whether a text was written by your lecturer.
SOLUTION:
Note: It is possible to have multiple answers, so long as the data are relevant, e.g.,
Handwriting samples (from lecturer, colleagues, friends)
Label: author of the text

Exercise 2: Model selection and search

(a) What is overfitting, and how can we reduce it?

SOLUTION:
Overfitting means fitting the training data closely, but not generalizing well. This can cause a model to fail to fit new data or predict future observations reliably.

Methods to reduce the overfitting:

- Increase size of training data.
- Reduce model complexity – via a regularization measure.
- Use a validation dataset to fine-tune the model. For example, use a validation dataset to prune a Decision Tree.
- Use cross-validation, where each data split is used as a validation dataset.
- Stop iterating when the performance metric stops improving.

(b) Suppose we have two models that fit the training data equally well, but one is less complex. Which do we prefer for predicting new (as yet unseen) data – the simpler model or the more complex model? Which of these models is more likely to overfit the data?

SOLUTION:
According to Ockham's Razor, we should use the simpler model. Minimum Message Length (MML, 1968), Akaike's Information Criterion (AIC, ca. 1970), Schwarz's Bayesian Information Criterion (BIC, 1978) and Minimum Description Length (MDL, 1978) are some attempts to put Ockham's razor into practice.

The simpler model is more likely to generalise well, and less likely to overfit the data.

(c) Suppose we have two models that are equally complex, but one fits the training data better than the other. Which do we prefer for predicting new (as yet unseen) data?

SOLUTION:
We should prefer the model which fits the training data better.

Note: There are no guarantees about which model will predict better, but, unless we have additional information to the contrary, there is no reason to prefer the model that has a worse fit with the training data.

(d) Explain how search can be used to assist Machine Learning.

SOLUTION:
All Machine Learning algorithms minimize or maximize some metric (e.g., RMSE for regression or Information Gain for decision trees). We therefore need a search algorithm to help find a good solution.

Exercise 3: Completeness, consistency and evaluation metrics

Consider the following information about two models that predict who will get a HD in FIT3080.

	Classes	
	HD	Not HD
Ground truth	A, B, C, D, E, F	G, H, I, J, K, L, M, N, O
Model A	A, B, C, D	**E, F**, G, H, I, J, K, L, M, N, O
Model B	A, B, C, D, E, F, **G**	H, I, J, K, L, M, N, O

(a) Determine the completeness and consistency of Model A and Model B for each class.

SOLUTION:
A definition is complete if it recognizes all instances of a class.
A definition is consistent if it does not classify any negative examples as instances of this class.

	Classes		Completeness and consistency	
	HD	Not HD	Class HD	Class Not HD
Ground truth	A, B, C, D, E, F	G, H, I, J, K, L, M, N, O		
Model A	A, B, C, D	**E, F**, G, H, I, J, K, L, M, N, O	incomplete consistent	complete inconsistent
Model B	A, B, C, D, E, F, **G**	H, I, J, K, L, M, N, O	complete inconsistent	incomplete consistent

(b) Can a model be incomplete and inconsistent? If so, give an example.

SOLUTION:

Model C: HD – A, B, C, D, G

(c) Calculate total accuracy for Model A and Model B, and per-class recall and precision for the HD and NotHD classes for Model A and Model B.

SOLUTION:

		Model A	**Model B**
Total accuracy	$\frac{\text{correct HD+correct NotHD}}{\text{all predictions}}$	$\frac{4+9}{15} = \frac{13}{15}$	$\frac{6+8}{15} = \frac{14}{15}$
Recall HD	$\frac{\text{predicted correct HD}}{\text{all HD}}$	$\frac{4}{6} = \frac{2}{3}$	$\frac{6}{6} = 1$
Precision HD	$\frac{\text{predicted correct HD}}{\text{all HD predictions}}$	$\frac{4}{4} = 1$	$\frac{6}{7}$
Recall NotHD	$\frac{\text{predicted correct NotHD}}{\text{all NotHD}}$	$\frac{9}{9} = 1$	$\frac{8}{9}$
Precision NotHD	$\frac{\text{predicted correct NotHD}}{\text{all NotHD predictions}}$	$\frac{9}{11}$	$\frac{8}{8} = 1$

(d) Can you posit a relationship between completeness/consistency and recall/precision?

SOLUTION:

A model that is complete with respect to a class C has recall = 1

A model that is consistent with respect to a class C has precision = 1

Exercise 4: Evaluation metrics usage: micro- and macro-averaging

Suppose you have a Machine Learning model that infers whether an elderly person has visitors in the house over 1 month (720 hours). The model has the following performance:

- Out of 80 hours when there were visitors, it detected 50.
- It also wrongly detected visitors in 20 hours when there weren't any visitors.
- It correctly detected no visitors in 620 hours.

Assuming you are measuring accuracy, which type of averaging would you use to assess the performance of this model? Why?

SOLUTION:

Let's first summarize the information:

Ground truth \ Model	Visitors	NoVisitors	Total ground truth
Visitors	50	30	80
NoVisitors	20	620	640
Total predicted	70	650	720

- Micro-averaged accuracy = $\frac{\text{Visitors correct} + \text{NoVisitors correct}}{\text{all predictions}} = \frac{50+620}{720} = 93\%$.

- Macro-averaged accuracy – we compute the accuracy for Visitors and NoVisitors separately. For binary classification, per-class accuracy is the same as recall, but for multi-class classification, we need to divide the dataset into two classes – the class of interest and all others.
  - Accuracy Visitors = $\frac{50}{80}$
  - Accuracy NoVisitors = $\frac{620}{640}$

  Macro-averaged accuracy = $\frac{1}{2}(\frac{50}{80} + \frac{620}{640}) = 79.7\%$.

The NoVisitors class is much larger than the Visitors class, and its results overwhelm those of the Visitors class, thus giving a false impression of better performance. So, in this case, macro-averaging is more informative.

Exercise 5: Decision trees

In a faraway galaxy, your spaceship has just landed on an alien planet, and your crew has begun investigating the local wildlife. Unfortunately, most of your scientific equipment is broken, so all you can tell about a given object is what colour it is, how many eyes it has, and whether or not it is alive. To make matters worse, none of you are biologists (let alone exobiologists), so you intend to use a Decision Tree to classify objects near your landing site as either alive or not alive. Use the table below to answer the following questions:

Object #	Colour	# of eyes	Alive
1	Red	4	Yes
2	Black	15	No
3	Red	40	Yes
4	Green	9	Yes
5	Black	25	No
6	Red	8	Yes
7	Black	9	Yes
8	Green	15	No

(a) What is the entropy of Alive?

SOLUTION:

$$H(X) = -\sum_{i=1}^{n} \Pr(x_i) \log(\Pr(x_i))$$

$$H(\text{Alive}) = -\Pr(\text{AliveYes}) \log(\Pr(\text{AliveYes})) - \Pr(\text{AliveNo}) \log(\Pr(\text{AliveNo}))$$

$$H(\text{Alive}) = -\frac{5}{8} \log\left(\frac{5}{8}\right) - \frac{3}{8} \log\left(\frac{3}{8}\right) = 0.9544$$

(b) What is the entropy of Alive after splitting on Colour?

SOLUTION:

The following table contains the per-colour information for being alive:

Colour	Alive	NotAlive	Total	Probability
Red	3	0	3	0.375
Black	1	2	3	0.375
Green	1	1	2	0.25
Total	5	3	8	1

$$H(\text{Alive}|\text{Red}) = 0$$

$$H(\text{Alive}|\text{Black}) = -\frac{1}{3}\log\left(\frac{1}{3}\right) - \frac{2}{3}\log\left(\frac{2}{3}\right) = 0.9183$$

$$H(\text{Alive}|\text{Green}) = -\frac{1}{2}\log\left(\frac{1}{2}\right) - \frac{1}{2}\log\left(\frac{1}{2}\right) = 1$$

$$H(\text{Alive}|\text{Colour}) = \Pr(\text{Red})H(\text{Alive}|\text{Red}) + \Pr(\text{Black})H(\text{Alive}|\text{Black}) + \Pr(\text{Green})H(\text{Alive}|\text{Green})$$

$$H(\text{Alive}|\text{Colour}) = 0.375 \times 0 + 0.375 \times 0.9183 + 0.25 \times 1 = 0.5944$$

(c) What is the Information Gain of splitting on Colour?

SOLUTION:

$$IG(\text{Colour}) = H(\text{Alive}) - H(\text{Alive}|\text{Colour})$$

$$IG(\text{Colour}) = 0.9544 - 0.5944 = 0.36$$

Exercise 6: Decision trees

You work for the ATO, and you want to build a Decision Tree to predict who will cheat on their taxes. Your historical data is in this table.

ID	Tax refund	Marital status	Taxable income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

(a) Calculate the Information Gain of splitting on the categorical features TaxRefund and MaritalStatus and the continuous feature TaxableIncome — for TaxableIncome, consider splitting on the mean and the median values. Select the feature with the highest Information Gain for your first split, and draw the resultant initial Decision Tree.

SOLUTION:

Q6

(a) Information Gain for each feature

H(Cheat) =	No	Yes		
	0.7	0.3	0.881290899	

H(TaxRefund)

TaxRefund	Cheat	NoCheat	Total	Probability
Yes	0	3	3	0.3
No	3	4	7	0.7
Total	**3**	**7**	**10**	**1**

H(Cheat | TaxRefundYes) =			0
H(Cheat | TaxRefundNo) =	0.428571429	0.571428571	0.985228136
H(TaxRefund)=	0.689659695		
IG(TaxRefund)=	**0.191631204**		

H(MaritalStatus)

MaritalStatus	Cheat	NoCheat	Total	Probability
Married	0	4	4	0.4
Single	2	2	4	0.4
Divorced	1	1	2	0.2
Total	**3**	**7**	**10**	**1**

H(Cheat | Married) =			0
H(Cheat | Single) =	0.5	0.5	1
H(Cheat | Divorced) =	0.5	0.5	1
H(MaritalStatus) =	0.6		
IG(MaritalStatus) =	**0.281290899**		

H(TaxableIncome) =

Split on MEAN **104**

TaxableIncome	Cheat	NoCheat	Total	Probability
>104	0	3	3	0.3
<=104	3	4	7	0.7
Total	**3**	**7**	**10**	**1**

IG(TaxableIncome split on MEAN) = 0.191631204 (same as TaxRefund)

Split on MEDIAN **92.5**

TaxableIncome	Cheat	NoCheat	Total	Probability
>92.5	1	4	5	0.5
<=92.5	3	2	5	0.5
Total	**4**	**6**	**10**	**1**

H(Cheat | TaxableIncome > 92.5) =	0.2	0.8	0.721928095
H(Cheat | TaxableIncome <= 92.5) =	0.6	0.4	0.970950594
H(TaxableIncome)=	0.846439345		
IG(TaxableIncome)=	**0.034851555**		

Therefore, the first split should be on MaritalStatus, yielding the following initial Decision Tree:

(b) Repeat the previous step for the second split, and draw the resultant Decision Tree.

SOLUTION:

(b) Information Gain for each feature after splitting on MaritalStatus

Single

	ID	Tax refund	Marital status	Taxable income	Cheat
	1	Yes	Single	125	No
	3	No	Single	70	No
	8	No	Single	85	Yes
	10	No	Single	90	Yes
			MEAN	**92.5**	
			MEDIAN	**87.5**	

H(Cheat) = No: 0.5, Yes: 0.5, ID: 1

H(TaxRefund)

TaxRefund	Cheat	NoCheat	Total	Probability
Yes	0	1	1	0.25
No	2	1	3	0.75
Total	**2**	**2**	**4**	**1**

H(Cheat | TaxRefundYes) =	0	1	0
H(Cheat | TaxRefundNo) =	0.666666667	0.333333333	0.918295834
H(TaxRefund)=	0.688721876		
IG(TaxRefund)=	**0.311278124**		

H(TaxableIncome)

Split on MEAN or MEDIAN **104**

TaxableIncome	Cheat	NoCheat	Total	Probability
Note that the mean and >87.5 or > 92.5	0	1	1	0.25
median have changed <=87.5 or <=92.5	2	1	3	0.75
Total	**2**	**2**	**4**	**1**

IG(TaxableIncome split on MEAN or MEDIAN) = 0.311278124 (same as TaxRefund)

We can continue splitting one more time, or use majority vote

Divorced

	ID	Tax refund	Marital status	Taxable income	Cheat
	5	No	Divorced	95	Yes
	7	Yes	Divorced	220	No
			MEAN	**157.5**	
			MEDIAN	**157.5**	

H(Cheat) = No: 0.5, Yes: 0.5, ID: 1

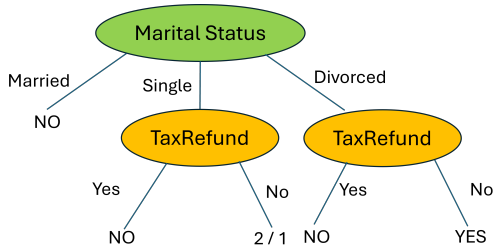
H(TaxRefund)

TaxRefund	Cheat	NoCheat	Total	Probability
Yes	0	1	1	0.5
No	1	0	1	0.5
Total	**1**	**1**	**2**	**1**

H(Cheat | TaxRefundYes) =	0	1	0
H(Cheat | TaxRefundNo) =	1	0	0
H(TaxRefund)=	0		
IG(TaxRefund)=	**1**		

IG(TaxableIncome split on MEAN or MEDIAN) = 0 (same as TaxRefund)

Therefore, the second split after Single or Divorced can be either TaxRefund or TaxableIncome, yielding the following second Decision Tree:



Notes:

- For TaxableIncome, the numbers in the second split are a subset of those in the first split, so the mean and median are different as well.
- When the samples in a leaf node have different classes, instead of performing more splits (which may overfit the data), you may decide to stop splitting. In this case, the class is determined by majority vote in the leaf (in this case, CheatYes for no TaxRefund).